



COBRA-WATCH

Cyber-Physical Surveillance Threat Intelligence & Detection Engine

STATUS: PRODUCTION-READY

309 / 309 TESTS PASSED (100%)

DPDP ACT 2023 COMPLIANT



1. Executive Summary (30-Second Presentation Pitch)

QUESTION	SIMPLE PRESENTATION ANSWER
What is the problem?	Normal CCTV systems blindly trust every camera feed. If a camera is hacked, unpatched, or fake, security guards waste critical time responding to false emergencies.
What is COBRA-WATCH?	A platform that calculates a Security Trust Score (0 to 100) for every camera <i>before</i> believing its physical video alerts.
How does it help?	High-trust alerts (80–100) trigger automated dispatch. Low-trust alerts (<50) are flagged for manual operator verification.



2. The 4-Step Pipeline

STEP	PIPELINE STAGE	TECHNOLOGY USED	WHAT IT DOES (1-SENTENCE SUMMARY)
Step 1	Camera Discovery	Shodan API + NVD CVEs	Finds exposed public IP cameras and checks if they carry known security bugs.
Step 2	Trust Score Engine	Custom Python Algorithm	Calculates 0–100 score based on password status, CVEs, age, and nearby camera proof.
Step 3	Video AI Pipeline	YOLOv8 + ByteTrack	Detects people/vehicles and checks if someone loiters (>10s) or crosses a boundary line.
Step 4	Live GIS Dashboard	React + Leaflet + WebSockets	Displays 2,369 sensors on a dark map with instant alert toast popups.

3. Technology Stack & Code Mapping

COMPONENT	TECHNOLOGY USED	SOURCE FILE LOCATION	IMPLEMENTATION & PURPOSE
User Interface	React 18 + Vite 5.4	<code>frontend/src/App.jsx</code>	Reactive single-page dashboard with collapsible side-panels.
Design System	Vanilla CSS3	<code>frontend/src/styles/globals.css</code>	Custom dark theme (#080b11), CRT scanlines, glassmorphism, keyframe animations.
Map Engine	Leaflet.js 1.9	<code>frontend/src/components/SurveillanceMap.jsx</code>	GPU-accelerated Leaflet map (<code>L.canvas()</code>) rendering 2,369 sensors at 60 FPS.
Satellite Basemap	Esri World Imagery	<code>frontend/src/utis/satelliteData.js</code>	High-resolution satellite aerial basemap tile layer toggle.
Backend API	Python 3.12 + FastAPI	<code>backend/main.py</code>	Asynchronous REST API server handling endpoints and WebSocket clients.
Database	SQLite 3 +aiosqlite	<code>backend/database.py</code>	Async database storing devices, OSINT news, alerts, and audit ledgers.
Real-Time Alerts	WebSockets Pub/Sub	<code>frontend/src/components/LiveAlerts.jsx</code>	Broadcasting instant threat detection event popups to connected clients.
Object Detection	YOLOv8 (ultralytics)	<code>video_pipeline/detector.py</code>	Neural network object detection identifying persons and vehicles from video.

COMPONENT	TECHNOLOGY USED	SOURCE FILE LOCATION	IMPLEMENTATION & PURPOSE
Motion Tracking	ByteTrack	video_pipeline/tracker.py	Multi-object motion tracking with Kalman filter state prediction and IoU matching.
Breach Math	2D Vector Math	video_pipeline/rules.py	Vector cross-product line breach detection and polygon ray-casting loitering time.
Trust Engine	Custom Algorithm	backend/services/trust_score_service.py	Weighted deduction model (0–100) calculating camera security risk.
Time-Decay Model	Exponential Decay	backend/routes/decay_router.py	Dynamic trust score erosion over time between active security vulnerability rescans.
Multi-Camera Fusion	Spatial-Temporal Fusion	backend/services/corroboration_service.py	Cross-camera event corroboration granting +20 trust bonuses to adjacent sensors.
Audit Ledger	SHA-256 Merkle Chain	backend/services/audit_ledger.py	Append-only cryptographic hash chain for non-repudiable audit logging.
Test Suite	Pytest (309 Passed)	backend/tests/	309-test master test suite verifying APIs, security, tracking, and breach rules.

4. Trust Score Points System (0–100 Scale)

SECURITY CHECK	POINTS EFFECT	WHY? (TECHNICAL JUSTIFICATION)
Base Starting Score	+100 Points	Perfect starting score for a brand-new camera.
No Password (Unauthenticated)	-30 Points	High risk! Anyone on the public internet can view or alter the stream.
Known Security Bugs (CVEs)	-10 Points per Bug	Camera has known unpatched NVD security vulnerabilities (max -40 points).
Unknown Owner / Org	-20 Points	Camera owner organization is missing or unverified.
Old Firmware (>1 year old)	-15 Points	Firmware has not been updated in over a year.
Confirmed by Nearby Camera	+20 Points Bonus	An adjacent camera (within 500m & 15 mins) also detected the physical event!

5. Reviewer Q&A Cheat Sheet

REVIEWER QUESTION	SIMPLE PRESENTATION ANSWER
"Why not just use YOLOv8 alone?"	"YOLO detects <i>what</i> is in the video, but can't tell if the camera is hacked. Our Trust Score checks camera security first so guards don't waste time on fake alerts."
"How do you check if cameras corroborate?"	"If Camera A alerts, we check if adjacent cameras within 500 meters also saw something within 15 minutes. If yes, +20 bonus!"
"Is this legal under Indian privacy laws?"**	"Yes! We only use public camera metadata via Shodan OSINT, run Video AI on authorized local footage, and log everything in a tamper-proof ledger."
"Is the system tested?"	"Yes, we ran a master suite of 309 automated tests covering APIs, security, tracking, and breach rules with a 100% pass rate. "

6. Presentation Demo Commands

1. Run Backend Server: `python backend/main.py` (`http://localhost:8000`)
2. Run GIS Dashboard: `cd frontend` then `npm run dev` (`http://localhost:5173`)
3. Run Master Test Suite: `pytest backend/tests video_pipeline/tests eval/test_eval_harness.py -v`